Credit Card Eligibility Cutoff Strategy: From Default Risk to a Decision

Personal portfolio project built on public data. It is not work experience and does not use any lender's data. All profit and cost figures are illustrative assumptions.

Code and documentation: https://github.com/nasimfn/credit-card-cutoff-strategy

The question and the answer

Where should a lender set a risk cutoff, what does each choice cost, and what would change the answer? The data is existing cardholders, so I framed the decision as eligibility for a credit limit increase: an eligible account that stays good earns 1 unit, and one that defaults costs X units. X is unknown, so every result is shown for X = 3, 5, 8 and 12.

Tier	Predicted default probability	Share of accounts	Actual default rate	Share of all defaults	Decision
A	below 7.7%	18.6%	4.8%	4.1%	Eligible under every cost ratio tested
B	7.7% to 25%	55.3%	15.1%	37.8%	Depends on the cost ratio: a business judgment
C	25% or more	26.1%	49.3%	58.1%	Not eligible under any cost ratio tested

For tier B, a simple break-even rule applies: make an account eligible while its predicted default probability is below 1/(1+X). It captured 95% to 99% of the best value found by simulation in all four scenarios. The single best cutoff is uncertain, because the value curves are flat near their peaks. What would change the answer: real revenue and loss figures, evidence of how limit increases change behaviour, and a population different from this 2005 Taiwanese sample.

How the project works, step by step

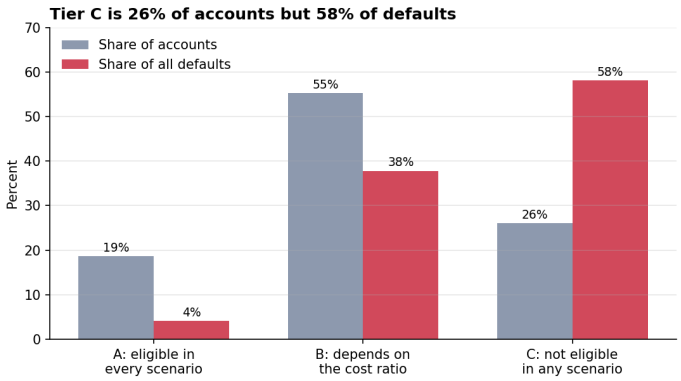
#	Step	What it does	Result
1	Download	Gets the public UCI Default of Credit Card Clients dataset (30,000 clients, Taiwan, 2005).	30,000 rows
2	Validate	Runs 22 data quality checks on the raw data and logs every failing record.	14 checks with findings
3	Clean	Applies documented decisions (undocumented codes, duplicates) and builds six features from account behaviour only. Sex, education, marital status and age are excluded from the model.	32-column clean file
4	Model	Trains logistic regression and gradient boosting, and compares them with one-variable baselines.	AUC 0.753 and 0.778 vs 0.683
5	Simulate strategy	Tests cutoffs under four cost ratios, checks the break-even rule, and bootstraps the uncertainty. Screens the cutoff for gaps by sex and age.	Three-tier rule
6	Query in SQL	Loads the scored accounts into SQLite and runs four queries: risk bands, cutoff table, tiers, segments.	4 result tables
7	Chart	Draws four charts, each stating its source and key caveat.	4 figures
8	Document and verify	Writes a data dictionary (32 and 13 fields) and a README, then reruns every script from a fresh clone of the repository.	Outputs identical

What the data and models showed

- **Data quality.** No missing values or duplicate IDs, but 345 education codes and 54 marital-status codes that the source does not define, and repayment codes (-2 and 0) that it does not define for most accounts. A one-month delay is common in September (3,688 accounts) but almost absent in earlier months. Accounts coded 'paid duly' default more (16.8%) than accounts coded 0 (12.8%). These were flagged, not corrected.
- **Models are good, not magic.** On 9,000 hold-out accounts, gradient boosting scored AUC 0.778 (cross-validation 0.785) and logistic regression 0.753, against 0.683 for a single repayment-status rule. Predicted and actual default rates agree to within about 3.5 percentage points in every risk band.
- **Concentration of risk.** Tier C is 26% of accounts but 58% of defaults; tier A is 19% of accounts and 4% of defaults.

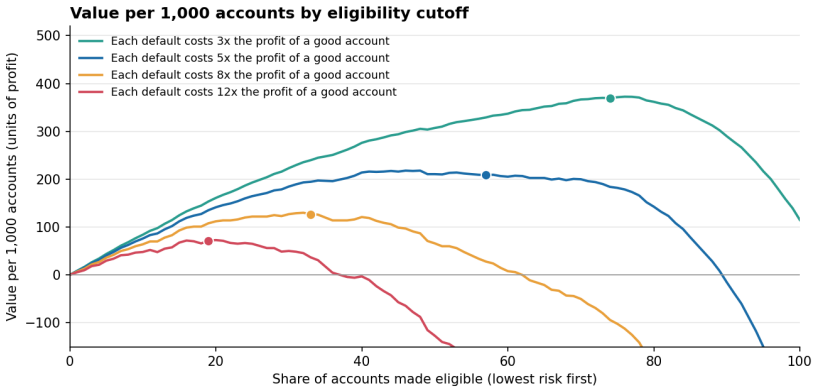
- **Screening check.** At the central cutoff, the share made eligible is 55.7% for men and 58.6% for women, and between 53.5% and 60.8% across age bands (lowest to highest ratio 0.88). Default rates also differ by group, so this is a screening check, not a fairness or legal test.

Chart: risk is concentrated in tier C



Default rates within tiers: A 4.8%, B 15.1%, C 49.3%.
Data: Default of Credit Card Clients (Yeh & Lien, 2009; UCI), 30,000 Taiwanese cardholders, Apr-Sep 2005. Results are on a 9,000-account hold-out set. The profit framing is illustrative and is not based on any lender's data.

Chart: value by cutoff under four cost ratios



The vertical axis is cut off at -150; curves keep falling beyond it. Dots mark the simple break-even rule (eligible while predicted default probability < 1/(1+X)). The curves are flat near their peaks, so the exact best cutoff is uncertain.
Data: Default of Credit Card Clients (Yeh & Lien, 2009; UCI), 30,000 Taiwanese cardholders, Apr-Sep 2005. Results are on a 9,000-account hold-out set. The profit framing is illustrative and is not based on any lender's data.

Limitations

- The data is from Taiwan in 2005, not Canadian cards. The cost ratio and the limit-increase framing are assumptions, not findings.
- About 1,991 hold-out defaults make cutoff estimates noisy; the bootstrap range for the best approval rate is wide (41% to 58% at X = 5).
- Excluding sex, education, marital status and age does not guarantee no indirect effect. This is not a regulatory-grade validation, and there is no time dimension to test stability.
- Source: Yeh, I. (2009). Default of Credit Card Clients. UCI Machine Learning Repository, doi.org/10.24432/C55S3H (CC BY 4.0). A CSV copy of the data was used, not the UCI file itself.

Tools: Python (pandas, scikit-learn, matplotlib), SQL (SQLite), Git. Developed with assistance from Claude (Anthropic) for drafting and reviewing code; the analytical decisions above were reviewed and chosen by me.